Supplementary material

Predictive reduced-order states for wall-pressure estimation of extreme vortex gust–airfoil interactions

S1. The transferred-forecasting ledger

The direct multi-horizon forecaster of §3.2.1 was built by auditing the TiRex forecasting line (Auer *et al.* 2025; Podest *et al.* 2026) ingredient by ingredient against this data. Adopted: the direct multi-horizon head, the quantile output, the per-window instance normalisation and the arcsinh transform, each of which survived ablation on validation. Not supported by validation for this data: future-known covariate conditioning; the oracle (G, D, Y) arm degrades the forecast in all three seeds (§4.4), overfitting held-out parameter combinations at the present case count. Rejected: capped-weight robustness in both forms tried, a saturating output nonlinearity that exploded during the gust transient and a loss-side weight cap that down-weighted exactly the transients the encounter is about. Deferred: the variate mixer and test-time adaptation, neither needed at this scale. The backbone bake-off at matched budget ties a hand-written exponential-gated cell (Beck *et al.* 2024) with the plain LSTM within seed noise; the plain cell at width 512 with nine quantiles is used throughout.

S2. Failure modes

Four mechanistic negatives complete the estimation story. First, filter consistency failure under the fixed criterion of §3.3.2 is a transformer-forecast phenomenon: the linear filter with encoded observations posts zero consistency failures where the transformer stack posts four, and under subsampled assimilation the transformer-forecast hybrid collapses already at every-second-frame updates while the linear filter degrades gracefully; every-frame pressure is necessary for the nonlinear filter. Second, the reconstructive baseline's estimatability failure in the dimension grid is an observation-side failure, verified three ways: linear probes on its true latents are healthy at every dimension and seed, the failure is insensitive to the observation map's ridge strength over three orders of magnitude, and the identical protocol succeeds for the linear and predictive families in every cell. Third, the ensemble smoother on the forecast-derived filter degrades it: at $N = 64$ the lagged cross-covariances are sampling noise, and the closed-form recursion on the linear stack is the working smoother. Fourth, the oracle-covariate negative of §S1 belongs to this list from the deployment side: more information, worse forecast, when the information indexes a family of held-out dynamics rather than the current state.

S3. Supplementary figures

The figures below support the Results and Discussion and are collected here to keep the main-text figure sequence focused on the narrative steps: the decoded vorticity fields behind the decode-similarity column of table 2 (figure S1) and the decodability-versus-estimability trade behind §5.3 (figure S2).

S4. The suited-operator forecast merit

This section is self-contained: the main text and appendix A now carry only the shared-operator comparison, and everything about the superseded protocol lives here.

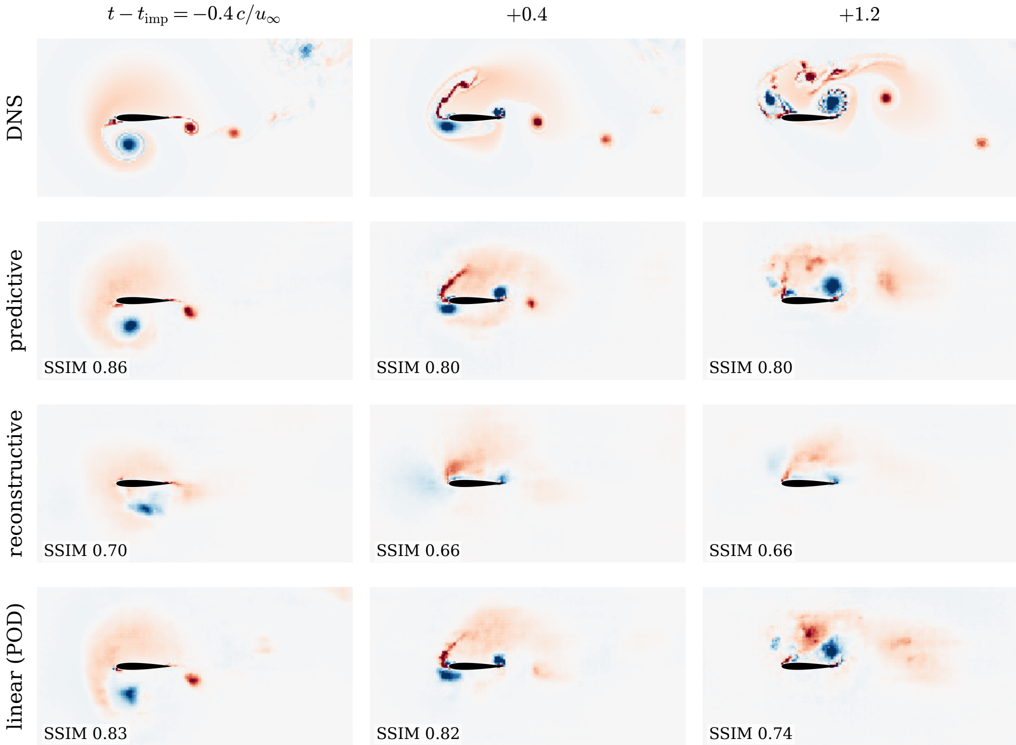

Figure S1. Decoded vorticity through the encounter on the representative held-out in-distribution test case ($|G| = 1.5$, the same encounter as figure 13). Rows: the simulation and the fields decoded from the encoded coefficient state of the predictive, reconstructive, and linear families; columns: a pre-impact, an impact, and a relaxation frame. Each decode is produced by the same diagnostic decoder that defines the decode-floor similarity of table 2, trained on the frozen encoder; the per-panel structural similarity is annotated. The pooled decode is soft by construction (§5.3); the comparison is between rows at matched frames, and no family's decode is part of any training loss.

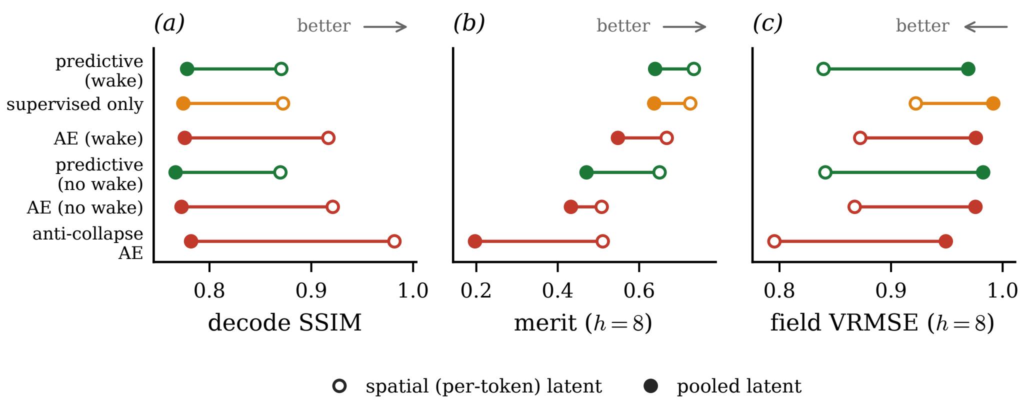

Figure S2. The decodability-versus-estimability trade of §5.3, quantified. Each panel connects the spatially resolved latent (open marker) to its pooled coefficient state (filled marker), per family, for the decode similarity, the matched-predictor merit, and the field forecast error. Pooling costs field fidelity in every family; the coefficient state is the price of a state a wall-pressure filter can track.

Table S1. Suited-operator forecast merit (mean over the five observables of the pooled nonlinear-probe R^2 at horizon eight) on the in-distribution test encounters, single fitted operator per family: matched transformer for the pooled controlled matrix, residual U-Net for the reference recipes. The wake-supervised predictive state's co-trained vector predictor reaches 0.755 under the same convention.

State	suited-operator merit
predictive (wake)	0.655
supervised only	0.621
AE (wake)	0.645
predictive (no wake)	0.294
AE (no wake)	0.400
anti-collapse AE	0.286
β-VAE	0.356
Fukami	−0.200
Fukami (wake)	0.203
POD	0.290

Two forecast operators can be fitted on a frozen latent, identically across families: an autoregressive transformer on the vector sequence, and a residual U-Net that tiles the vector to a grid and forecasts it convolutionally. They do not have the same stability. The transformer is stable across the pooled states of the controlled matrix, but refitted with the same recipe and budget on the reference latents, the β-VAE and the two Fukami autoencoders, which are plain convolutional autoencoders without skip connections, it diverges to large negative merit. The residual U-Net is stable on every latent. Before the shared-operator recomputation the families table therefore scored each state under whichever of the two was numerically stable on it, transformer for the pooled states and U-Net for the reference latents, which is the suited-operator protocol. Its values, at horizon eight, are retained here because the choice of operator is itself informative: the no-wake predictive state that collapses under the shared direct multi-horizon forecaster (−0.083 at the same horizon) posts 0.294 under its suited transformer, and the reference recipes reorder. The shared-operator column of the main text (table 2) is the primary comparison; this table documents the superseded protocol.

References

Auer, Andreas, Podest, Patrick, Klotz, Daniel, Böck, Sebastian, Klambauer, Günter & Hochreiter, Sepp 2025 TiRex: zero-shot forecasting across long and short horizons with enhanced in-context learning. *arXiv preprint arXiv:2505.23719* .

Beck, Maximilian, Pöppel, Korbinian, Spanring, Markus, Auer, Andreas, Prudnikova, Oleksandra, Kopp, Michael, Klambauer, Günter, Brandstetter, Johannes & Hochreiter, Sepp 2024 xLSTM: extended long short-term memory. *arXiv preprint arXiv:2405.04517* .

Podest, Patrick, Pichler, Marco, Bürger, Elias, Zólyomi, Levente, Voggenberger, Bernhard, Berghammer, Wilhelm, Klotz, Daniel, Böck, Sebastian, Klambauer, Günter & Hochreiter, Sepp 2026 TiRex-2: generalizing TiRex to multivariate data and streaming. *arXiv preprint arXiv:2607.01204* .